BITS Pilani

Introduction to AI Systems

Dr. Minati Rath
mrath@wilp.bits-Pilani.ac.in

innovate achieve lead

BITS Pilani

<SS ZG662, Introduction to AI Systems >
Session 2

These slides are prepared by the instructor, with grateful acknowledgement of many others who made their course materials freely available online.

Session Content

Topics: Categories of AI Systems

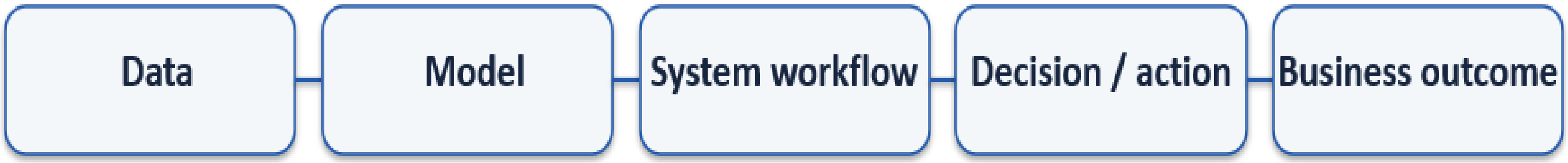
- Predictive AI
- Generative AI
- Recommender Systems
- Conversational AI
- Computer Vision
- Autonomous Systems
- Case Studies: ChatGPT, Netflix Recommendations, Autonomous Vehicles

AI Systems: From Model to Business Outcome

Corporate AI value comes from connecting a model to a business process, decision, user, or physical action.

A high-performing model can still produce poor business outcomes if the workflow, data, incentives, or controls are wrong.

Think in layers: data → model → system workflow → user/action → measurable outcome.



Choosing the Right AI System for a Business Problem

Forecast an outcome? → Predictive AI.

Create new content? → Generative AI.

Select the most relevant item? → Recommender System.

Interact through natural language? → Conversational AI.

Interpret images or video? → Computer Vision.

Sense, decide and act in the environment? → Autonomous System.

Six AI Categories — Comparative View

Predictive AI

Estimate a future/unknown outcome

Generative AI

Create new content

Recommender Systems

Rank/select relevant items

Conversational AI

Understand and respond in dialogue

Computer Vision

Extract meaning from visual data

Autonomous Systems

Perceive, decide and act

Predictive AI

Forecasting and decision support for measurable business outcomes

Predictive AI

Predictive AI uses historical and real-time data to estimate future outcomes, risks, or behaviors.

Models may combine structured business data, time-series signals, external variables, and engineered or learned features.

Structured business data – Customer age, account balance, product category, or store location from ERP/CRM systems.

Time-series signals – Daily sales volume, hourly website traffic, or monthly energy consumption over time.

External variables – Weather conditions, inflation rate, public holidays, or competitor pricing data.

Engineered features – 30-day average sales, customer lifetime value (CLV), or purchase frequency calculated from raw data.

Learned features – Embeddings generated by a neural network representing customer behavior, product descriptions, or images

The system should be evaluated not just on prediction accuracy, but on business impact, uncertainty, robustness, and the cost of prediction errors.

Data ⟶ Features ⟶ Model ⟶ Prediction ⟶ Decision

Predictive AI

Example: **customer churn** — usage frequency, tenure, support contacts, payment behaviour.

Model output can be probability of churn rather than a simple yes/no label.

Business layer converts probability into an intervention: retain, monitor, or do nothing.

Prediction quality should be assessed at the point where the business actually acts.

📊 Signals	🧠 Prediction	🎯 Risk	⚡ Action
Usage ↓ 40%	Learns patterns	**82% churn risk**	**Retain**
3 complaints	Historical data	High	Monitor
Late payments	Customer profile	Medium / Low	No action
Competitor offer	→ Model		

If you can intervene with only 2,000 customers, which 2,000 should the model prioritize?

Predictive AI – Uplift Modelling

Ordinary churn model asks the wrong question for intervention

A conventional model learns: **“Who is likely to churn?”**

For example: **Customer A → 90% churn probability**

But this doesn't tell us whether a *retention campaign* will actually work.

A customer may be almost certain to churn **regardless of what we do**.

Uplift modeling asks a different question

An uplift model asks: **“*How much will the outcome change if we intervene?*”**

Conceptually: **Uplift = P(stay | intervention) − P(stay | no intervention)**

Customer	Without Intervention	With Intervention	Effect of Intervention	Customer Type	Business Action
A	95% churn	90% churn	**Very Low**	**Lost Cause**	Do not prioritize
B	80% churn	20% churn	**Very High**	**Persuadable**	**Prioritize**
C	5% churn	3% churn	**Very Low**	**Sure Thing**	No intervention needed
D	20% churn	60% churn	**Negative**	**Do Not Disturb**	Avoid intervention

*The interesting customer is **B**. The churn model might identify A and B as high-risk, but the uplift model tells us that **B is much more persuadable***

Predictive AI – Uplift Modelling

The best customer to target is not necessarily the one with the highest churn risk — it is the one whose behaviour can be positively changed by the intervention

Customer Data → Churn Model → Uplift Model → Customer Value + Intervention Cost → Optimal Action / Targeting

Who is likely to churn? → Whose behaviour can we influence? → Is intervention economically worthwhile? → Who should we target, and how?

Predictive AI — Real Business Use Cases

Finance: credit risk, fraud detection, cash-flow forecasting.

Retail: demand forecasting, inventory planning, customer churn.

Manufacturing: predictive maintenance and quality prediction.

Healthcare: risk stratification and readmission prediction.

Operations: capacity, delivery time, staffing, and resource demand.

Predictive AI — Metrics and Business KPIs

Model metrics answer: how accurate is the prediction?

Business KPIs answer: did the prediction improve the decision?

Example: fraud model → recall and false-positive rate → prevented loss and investigation workload.

Example: demand forecast → MAE/MAPE → stockouts, inventory cost, and service level.

Always connect model evaluation to operational economics.

Predictive AI — Deployment Challenges

Data drift: future input patterns change. Suppose a bank's model was trained using customer data from 2022–24. It learned patterns from: Salary, credit-card usage, loan amount, transaction behaviour
Now in 2026, customers behave differently because of economic conditions, digital payment adoption, or new products.
The **input distributions have changed**, even if the meaning of the variables has not.
Example: Average monthly digital transactions were ₹20K during training but are now ₹60K.
Concept drift: the relationship between inputs and outcomes changes. Suppose historically high credit-card utilization → high probability of default. But economic conditions change, and customers with high utilization are now mostly affluent customers who pay their bills regularly. The input hasn't necessarily changed dramatically. What changed is the relationship between the input and the outcome. So the model's old relationship is no longer reliable.

Predictive AI — Deployment Challenges

Label leakage: training uses information unavailable at prediction time. Suppose we want to predict:Will this customer default next month? But one training feature is: Number of collection calls made after the customer missed payment. That feature may be highly predictive of default—but it is available only after the problem has already occurred. The model appears extremely accurate during training/testing, but cannot use that information when making a real prediction.

Threshold design: false positives and false negatives have different costs.

Monitoring must track both model behaviour and business outcomes.

Monitoring must track both model behaviour and business outcomes.

Classification vs Regression in Business

CLASSIFICATION

Output: category or probability
Examples: fraud / not fraud; churn / retain; defect / acceptable
Typical metrics: precision, recall, F1, ROC-AUC

REGRESSION

Output: numeric value
Examples: demand, revenue, delivery time, energy consumption
Typical metrics: MAE, RMSE, MAPE

Predictive AI — When to Use / When Not to Use

USE WHEN

Historical signals exist.
A target or measurable outcome is clear.
Predictions can change a decision.
The decision cycle repeats.

RETHINK WHEN

No reliable target exists.
The process is fundamentally creative.
Actions cannot change after prediction.
Data is too sparse or unstable.

Generative AI

Content creation, reasoning support and enterprise knowledge workflows

Generative AI — Enterprise Workflow

Enterprise GenAI usually combines a foundation model with business context and controls.

Context may come from documents, databases, APIs, user history, or tools.

The system should evaluate not just language quality but task correctness and grounding.

Bank - 1. Foundation model - The LLM provides the basic language and reasoning capability. 2. Business context - The model is given access to the bank's relevant information:

Documents → Policies → Customer data → APIs → Internal systems

An employee asks:“***Can this customer get a fee waiver***?”

The model shouldn't answer from its general training. It retrieves the bank's current fee policy, checks the customer's account information, and possibly calls an internal API.

User task → **Prompt + context** → **Foundation model** → **Tools / retrieval** → **Output** → **Evaluation**

Generative AI — Enterprise Workflow

3. Controls - The system must enforce enterprise rules:Is the employee authorized to see this customer data?Which information can the model access?Can it actually perform the requested action?Should the response require human approval?

So the architecture is: *Foundation Model → Business Context → Enterprise Controls → Business Response*

Why evaluation is differentA response can sound perfectly fluent but still be wrong. For example:“Yes, the customer qualifies for a fee waiver.”The language may be excellent, but if the current policy says the customer doesn't qualify, the system has failed.So we evaluate:Language quality + Task correctness + Grounding + Reliability + Business outcome

Enterprise GenAI is not just about generating good language - *it is about generating the right answer using the right business context under the right controls.*

RAG, Fine-Tuning and Prompt Engineering

PROMPT ENGINEERING

Change instructions and examples.
Fastest to iterate.
Useful for behaviour, format and task framing.

RAG

Retrieve relevant enterprise knowledge at runtime.
Useful when facts change or documents are private.

FINE-TUNING

Adapt model behaviour style, patterns, or tasks.

Generative AI Across Enterprise Functions

Customer service: response drafting, conversation summaries, knowledge assistance.

HR: policy Q&A, job-description drafting, employee self-service.

Finance: document extraction, commentary, reporting support.

Legal: clause comparison, document review support, research assistance.

Engineering: code generation, testing, documentation, incident support.

Generative AI - HR Policy Assistant

User task"*Can I work remotely for 3 weeks from another country*?"→ Prompt + Context (Employee role, location, tenure, and the relevant question are included)→ Foundation Model
LLM understands the request and reasons over the provided context.→ Tools / Retrieval - Retrieves the latest remote-work policy from the company knowledge base and, if needed, checks employee information through an HR API.→ Output "Your role is eligible for remote work, but working from another country for 3 weeks requires manager and HR approval."→ Evaluation Check - whether the answer: follows the current policy uses the correct employee information answers the actual question provides a grounded response rather than inventing a rule.
Employee Question → Prompt + Employee Context → Foundation Model → Policy Retrieval + HR API → Decision / Response → Correctness + Grounding + Policy Compliance

Generative AI — Productivity vs Automation

Productivity: AI assists a human who remains responsible for the workflow. Ex **-AI assists the employee-** A customer-support executive asks AI: **“Summarize this customer's complaint and suggest a response.”** AI produces the summary and draft. **Human → reviews → sends response** The human remains responsible for the decision.

So an incorrect AI response is caught by the employee.

Automation: AI output directly triggers a downstream system action.

Moving from productivity to automation increases the need for validation, permissions, audit trails, and failure recovery.

Enterprise maturity is therefore a workflow question, not simply a model question.

Now suppose the system receives a customer complaint and automatically: **Reads complaint → determines issue → approves refund → triggers payment.** There is no employee reviewing every case.

Generative AI — Hallucination, Bias and Data Risks

Hallucination: plausible content that is unsupported or incorrect.

Confidentiality: sensitive information may enter prompts, context, logs, or outputs.

Bias: generated content can reproduce patterns in training data.

Prompt injection: untrusted content can attempt to manipulate system behaviour.

Controls: retrieval, access control, output validation, red teaming, logging, and human review.

Generative AI — Enterprise Implementation Considerations

Choose the model based on task quality, latency, cost, context length, and deployment constraints.

Define an evaluation set before optimisation.

Separate model permissions from application permissions.

Track cost per task, failure rate, user adoption, and task completion.

Design escalation paths for uncertain or high-impact requests.

Recommendation Engine — End-to-End Architecture

The core problem is ranking: which items deserve attention first?

Most production systems use a multi-stage architecture to balance scale and model complexity.

Candidate Generation ⟶ Ranking ⟶ Top-N

CANDIDATE GENERATION

Reduce millions of items to hundreds.
Fast retrieval models and similarity signals.

RANKING

Use richer user, item and context features.
Predict relevance or expected utility.

TOP-N SERVING

Apply diversity, business rules and constraints.
Return a small set for the user.

Recommender Systems

Ranking the right item, content or action for the right user and context

Recommendation Metrics — CTR, Conversion, NDCG, Revenue

CTR measures interaction, not necessarily satisfaction.

Conversion measures whether recommendations contribute to the desired action.

NDCG rewards placing highly relevant items near the top of the list.

Revenue and retention connect recommendation quality to business value.

Also monitor diversity, coverage, novelty, and fairness.

Personalisation Signals

User signals: history, recency, frequency, preferences, purchases.

Item signals: category, price, popularity, freshness, semantic representation.

Context signals: time, device, location, session intent, current task.

Business signals: inventory, margin, campaigns, contractual constraints.

Good systems distinguish short-term intent from long-term preference.

Cold Start, Bias and Feedback Loops

Cold start: little information about a new user or item.

Popularity bias: already-popular items receive more exposure and more interaction.

Feedback loop: recommendations affect behaviour, which becomes future training data.

Exploration can discover new preferences and reduce over-specialisation.

Governance should consider exposure and user outcomes—not only clicks.

Natural-language interaction connected to enterprise knowledge and actions

Enterprise Conversational AI Architecture

Modern conversational systems combine language understanding with context, retrieval, tools, and policy.

The interface may be chat, voice, messaging, or embedded workflow interaction.

User

Language model

Context / memory

Knowledge / tools

Response / action

Intent ⟶ Context ⟶ Retrieval ⟶ Response ⟶ Action

Intent: infer what the user is trying to accomplish.

Context: preserve the information required to continue the task.

Retrieval: access authoritative enterprise information.

Response: communicate clearly and at the appropriate level.

Action: update systems, create tickets, place orders, or trigger workflows when authorised.

LLM Chatbots vs Traditional Chatbots

TRADITIONAL

Intent classification + predefined flows
Strong control and predictable paths
Limited flexibility with unseen requests
Often efficient for narrow tasks

LLM-BASED

Flexible natural-language interaction
Can summarise, reason over context and generate responses
Higher variability
Requires stronger grounding, evaluation and guardrails

Conversational AI — Customer Service and Employee Support

Customer service: answer, triage, summarise, authenticate, and route.

Employee support: policy questions, IT troubleshooting, HR self-service.

Sales: product discovery, qualification, proposal support.

Operations: natural-language access to dashboards and enterprise data.

Value comes from reducing resolution time and improving task completion—not simply producing fluent replies.

Conversational AI — Risks, Escalation and Human-in-the-Loop

Escalate when confidence is low, policy requires human judgement, or the user requests a human.

Prevent unauthorised actions through identity, access control and tool-level permissions.

Log relevant events without exposing unnecessary sensitive data.

Measure containment, resolution, escalation quality, customer effort, and error severity.

Turning visual data into operational signals

Computer Vision — Enterprise Pipeline

Camera or image source → preprocessing → vision model → structured output → business decision.

Image / video | Preprocess | Vision model | | Business rule | Action

segmentation, OCR, tracking, or visual embeddings.

Image Classification vs Detection vs Segmentation

CLASSIFICATION

What is present?
Example: defective / acceptable product.
One or more labels.

DETECTION

What objects are present and where?
Example: locate defects, vehicles or people.
Bounding boxes + labels.

SEGMENTATION

Which pixels belong to each region?
Example: precise defect area or medical structure.
Pixel-level masks.

Vision AI in Manufacturing and Quality Inspection

Inspect surface defects, assembly errors, dimensions, labels, and packaging.

Integrate camera stations with PLCs, MES, quality systems, and operator workflows.

Business value depends on false-reject rate, missed-defect rate, throughput, and rework cost.

Model accuracy alone does not capture the cost of inspection errors.

Vision AI in Retail, Healthcare and Logistics

innovate achieve lead

Retail: shelf availability, visual search, checkout support.

Healthcare: image analysis and workflow prioritisation under clinical oversight.

Logistics: parcel recognition, barcode/OCR, damage detection, warehouse perception.

Each domain has different validation, privacy, safety, and integration requirements.

Vision AI — Deployment Challenges

Camera placement and image quality can dominate model performance.

Edge vs cloud deployment affects latency, bandwidth, privacy, and maintenance.

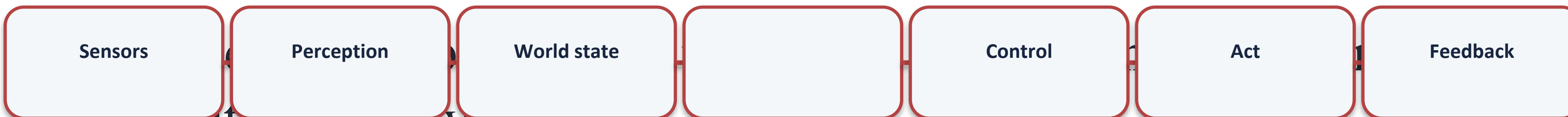
Rare defects make representative evaluation difficult.

Model updates must be tested against historical failure modes.

Human escalation should exist for uncertain or ambiguous cases.

Autonomous Systems — Perception ⟶ Planning ⟶ Action

Autonomous systems continuously observe their environment and choose actions toward a goal.



and future observations.

Safety constraints and fallback behaviour are part of the system architecture.

Sensor Fusion and Environment Perception

Different sensors provide complementary information.

Autonomous vehicles may combine cameras, radar, lidar, GPS, maps, and inertial measurements.

Fusion reduces ambiguity but introduces calibration, synchronisation, and failure-management requirements.

Perception must estimate not only what is present but where it is and how it may evolve.

Decision-Making, Planning and Control

Prediction estimates how agents and objects may move.

Planning selects a safe and useful trajectory or sequence of actions.

Control converts the plan into physical commands.

Fast feedback is required because the environment changes continuously.

Testing must cover rare, adversarial, and unexpected situations.

Levels of Autonomy

Assistive: AI provides information or recommendations.

Supervised autonomy: system acts within a bounded scope with human oversight.

Conditional autonomy: system handles a defined operational domain with limited intervention.

High autonomy: system manages a broad set of situations but still depends on defined operating conditions.

Autonomy should always be specified by task, environment, and fallback responsibility.

Autonomous Systems — Safety and Failure Handling

Define safe operating boundaries before deployment.

Use redundancy and fallback strategies for sensor, compute, communication, and model failures.

Monitor uncertainty and detect when the system is outside its validated operating domain.

Fail-safe behaviour should be designed, tested, and auditable.

Simulation and scenario-based testing complement real-world testing.

Autonomous Systems

Perception, prediction, planning and control in a closed loop

ChatGPT — Generative + Conversational + Tool Use

Generative: creates text, code, summaries, plans, and other outputs.

Conversational: maintains interaction context and supports iterative task completion.

Tool use: can extend the model with external capabilities where the deployment permits it.

These capabilities can be combined into an enterprise workflow rather than used as isolated features.

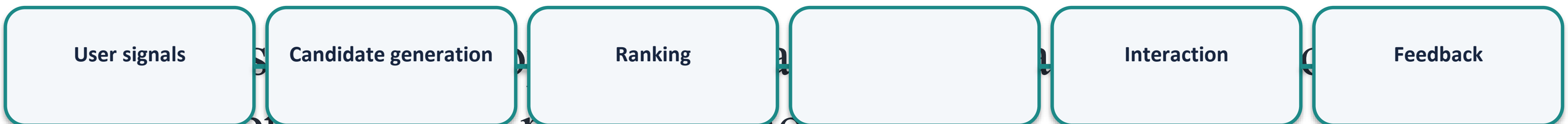
Executive takeaway

Use the category to frame the business problem; use the architecture to

- Predictive AI estimates what may happen.
- Generative AI creates new content or structured outputs.
- Recommender Systems decide what deserves attention.
- Conversational AI provides a natural-language interface to knowledge and actions.
- Computer Vision turns visual data into operational signals.
- Autonomous Systems close the loop by deciding and acting.
- The strongest enterprise systems combine categories around a measurable business outcome.

Netflix — Recommendation System Architecture

User interaction data and contextual signals feed candidate generation and ranking.



small set of highly relevant choices.

Recommendation surfaces can include rows, titles, ordering, and artwork or presentation choices.

Netflix — Personalisation, Ranking and Feedback

Personalisation uses historical behaviour together with current context.

Ranking determines which content gets attention and in what order.

Feedback is continuous: viewing and interaction behaviour becomes new evidence.

Exploration, diversity, freshness, and long-term engagement must be balanced against short-term clicks.

Netflix — Business Value and Recommendation Challenges

Value: reduce search effort, increase content discovery, improve engagement and retention.

Challenge: popularity bias can narrow exposure.

Challenge: cold-start content has limited interaction history.

Challenge: recommendation quality can be affected by changing user intent.

Key lesson: recommendation is a continuously optimised business system, not a one-time model.

Case Study: ChatGPT

Generative AI + Conversational AI + tools

Perception ⟶ Prediction ⟶ Planning ⟶ Control

Perception asks: what is around me?

Prediction asks: what might other road users do next?

Planning asks: what should the vehicle do?

Control asks: how should the vehicle execute that plan?

Feedback asks: did the environment respond as expected?

Three Case Studies — What They Teach Us About AI Systems

CHATGPT

Primary categories:
Generative AI
Conversational AI

Lesson: model capability becomes valuable when connected

NETFLIX

Primary category:
Recommender Systems

Lesson: ranking + feedback + experimentation turn predictions into personalised experiences.

AUTONOMOUS VEHICLES

Primary categories:
Computer Vision
Predictive AI
Autonomous Systems

Lesson: closed-loop action demands safety, perception, planning